					Course Code:A2610001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		ENGLISH for TECHNICAL COMMUNICATION			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Identify any four expressions for asking for permissions.	**1**	**2**	**2**
	b).	Complete the sentences with the appropriate words as directed in the brackets. i. The students are interested ___ learning new technologies. (Preposition) ii. The success of _____ project depends on effective teamwork. (Article)	**2**	**3**	**2**
	c).	Construct one example sentence for any two of the following idiomatic expressions. i. A piece of cake ii. Break the ice iii. Once in a blue moon iv. Spill the beans	**3**	**3**	**2**
	d).	Write about the use of skimming technique in reading the texts.	**4**	**3**	**2**
	e).	Dos and don'ts of social media messaging.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Develop an essay on the strategies that can help a person sound smarter, more confident, and articulate when communicating.	**1**	**3**	**5**
	b).	Write a dialogue between a student and a librarian about borrowing a reference book and renewing the library membership.	**1**	**3**	**5**
		OR			
3.		Construct a paragraph on "The Importance of Time Management" in about 100–120 words and list out the cohesive devices used in the paragraph.	**1**	**3**	**10**
		UNIT-2			
4.	**a).**	Explain how eco-conscious negotiation strategies can be applied to address an environmental issue and promote a sustainable future.	**2**	**3**	**5**

	b).	Employ appropriate verb forms to fill in the blanks. i. By next year, she ___ (finish) her degree. ii. The train ___ (arrive) at the station just now. iii. I ___ (study) English for three years. iv. When I called him, he ___ (sleep). v. Listen! Someone ___ (play) the guitar.	**2**	**3**	**5**
		OR			
5.		Apply appropriate grammatical rules and correct any **TEN** of the following sentences. i. Each of the engineers are responsible for testing the software. ii. The equipment in the laboratory were not functioning properly. iii. He discussed about the advantages of artificial intelligence. iv. The results of the experiment shows a significant improvement. v. The engineer suggested to use a different algorithm for the project. vi. If the system will fail, the backup server will be activated. vii. He is resembling his father. viii. The film show began before we reached the theatre. ix. Neither the project manager nor the team members was aware of the technical issue. x. She is an university student pursuing engineering. xi. Rahul went to school despite of having a fever. xii. I prefer chocolate milkshake than vanilla ice cream.	**2**	**3**	**10**
		UNIT-3			
6.	**a).**	Illustrate the appropriate ethical guidelines to address a challenge arising from the use of artificial intelligence or other disruptive technologies.	**3**	**3**	**5**
	b).	Construct sentences by substituting the given phrases with suitable one-word for any **FIVE** of the following. i. One who loves mankind. ii. One who cannot be defeated iii. Hobby of collecting coins. iv. Something that cannot be avoided v. A speech made without preparation vi. A person who spends money wastefully vii. A person who designs and develops software viii. A person who loves and collects	**3**	**3**	**5**
		OR			
7.	**a).**	Construct a formal letter to the Editor of a newspaper expressing your concern about the increasing use of plastic in your city. Apply appropriate letter-writing conventions and suggest possible solutions.	**3**	**3**	**10**

		UNIT-4			
8.	**a).**	Develop an essay on how innovative business practices can be applied to address challenges faced by modern startups.	**4**	**3**	**5**
	b).	Analyze the following hints to construct a formal email by applying appropriate writing techniques, tone, and structure. **Student ---> HR Manager:** New training/internship program -latest technology- useful – my career-want to be included- participate keenly-learn and implement- benefit of the company.	**4**	**3**	**5**
		OR			
9.	**a).**	Analyze the given passage and write answers to the following questions. Technology has transformed the way students learn and access information. Digital platforms, online courses, and educational applications allow learners to study at their own pace and explore subjects beyond the classroom. Engineering students, in particular, can use simulations, virtual laboratories, and collaborative tools to understand complex concepts more effectively. However, technology should complement traditional learning rather than completely replace it. Excessive dependence on digital resources may reduce face-to-face interaction and critical thinking. Therefore, students should use technology responsibly and balance digital learning with classroom discussions, practical activities, and independent study. Questions i. How has technology transformed the learning process? ii. What are the advantages of digital platforms and online courses? iii. How can engineering students benefit from simulations and virtual laboratories? iv. Why should technology not completely replace traditional learning? v. What are the possible disadvantages of excessive dependence on digital resources? vi. What does the word “complement” mean in the context of the passage? vii. Find a word from the passage that means “working together.” viii. Find a word from the passage that means “too much or more than necessary.”	**4**	**3**	**10**

		ix. What three activities does the passage recommend balancing with digital learning? x. What is the central idea of the passage?			
		UNIT-5			
10.	**a).**	Demonstrate how core values can be integrated into ethical decision-making and socially responsible leadership.	**5**	**3**	**5**
	b).	Construct the following sentences as directed in the brackets. i. She has written many novels. (Add Question Tag) ii. The report was prepared by the committee. (Convert it into Active Voice) iii. He worked hard to pass the examination. (Convert it into a Complex Sentence) iv. Very few engineering colleges in India are as reputed as IIT Madras. (Convert it into the Superlative Degree) v. Smrithi is the cleverest girl in the class. (Change it into Comparative Degree)	**5**	**3**	**5**
		OR			
11.	**a).**	Develop the content to be used in a blog on 'Social Media'.	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:A2610002

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R26**

I B.Tech. I Semester MODEL QUESTION PAPER

LINEAR ALGEBRA & CALCULUS

(Common to All Branches of Engineering)

Time: 3 Hrs. **Max. Marks: 60 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

5 x 2 = 10 Marks

			CO	KL	M
1.	(a)	Find the value of k such that rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2	1	3	2
	(b)	Find the singular values of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	2	3	2
	(c)	Find the value of c in [0, 4] for $f(x) = x(x-2)(x-3)$ by using Lagrange's mean value theorem.	3	3	2
	(d)	Evaluate $\underset{\substack{x\to\infty \\ y\to 2}}{Lt}\left(\frac{xy+1}{x^2+2y^2}\right)$	4	3	2
	(e)	Evaluate $\int_0^2 \int_0^4 (x^2+y^2)\,dxdy$	5	3	2

5 x 10 = 50 Marks

		UNIT-1			
2.	(a)	Determine the rank of the matrix $\begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$ by reducing it to echelon form.	1	3	5
	(b)	Find the inverse of the matrix $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ by using Gauss – Jordan method.	1	3	5
		OR			
3.	(a)	For what values of a and b do the equations $x + 2y + 3z = 6$, $x + 3y + 5z = 9$ and $2x + 5y + az = b$ have (i) No solution (ii) An unique solution (ii) Infinitely many solutions	1	3	5
	(b)	Solve the equations $20x + y - 2z = 17$, $3x + 20y - z = -18$and $2x - 3y + 20z = 25$ by Gauss-Seidel iteration method.	1	3	5
		UNIT-2			

4	Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$		2	3	10
	OR				
5.	Reduce $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ to sum of squares by linear transformation and hence find the matrix of the transformation, Nature, Signature and index of the quadratic form.		2	3	10
		UNIT-3			
6.	(a)	Verify Rolle's theorem for $\frac{\sin x}{e^x}$ in $(0, \pi)$.	3	3	5
	(b)	Expand $f(x) = \cos x$ by Maclaurin's theorem with Lagrange's form of remainder.	3	3	5
		OR			
7.	(a)	Find the appropriate value of c used in Cauchy's Mean-Value theorem for $f(x) = \ln x$ in [1, e]	3	3	5
	(b)	Expand $\tan^{-1}x$ in powers of $(x-1)$ upto fourth degree term.	3	3	5
		UNIT-4			
8.	(a)	If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$	4	3	5
	(b)	Expand $f(x,y) = e^y \log(1+x)$ in powers of x and y using Maclaurin's series.	4	3	5
		OR			
9.	(a)	If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	4	3	5
	(b)	Discuss the maxima and minima of $f(x,y) = x^3y^2(1-x-y)$	4	3	5
		UNIT-5			
10.	(a)	Evaluate $\int\int xydxdy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$ by changing it in to polar coordinates	5	3	5
	(b)	By the change of order of integration evaluate $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dydx$.	5	3	5
		OR			
11.		Evaluate $\iiint_V \frac{dxdydz}{x^2+y^2+z^2}$ where V is the volume of the sphere $x^2 + y^2 + z^2 = a^2$	5	3	10

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2610401
		SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE(A)			**R26**
		I B.Tech I Semester MODEL QUESTION PAPER			
		APPLIED PHYSICS			
		(for ECE)			
Time: 3Hrs.					**Max. Marks: 60M**
		Answer Question No.1compulsorily			
		Answer **ONE** Question from **EACH UNIT**			
		Assume suitable data if necessary			
					5x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a)**	Explain any four conditions for the formation of an interference pattern.	1	2	2
	b)	Discuss the applications of Laser.	2	2	2
	c)	Describe the drift and diffusion currents in semiconductors.	3	2	2
	d)	Explain the de Broglie concept of matter waves.	4	2	2
	e)	Describe the divergence and curl of an electric field.	5	2	2
					5x 10 = 50 Marks
		UNIT-1	**CO**	**KL**	**M**
2.		Describe the formation of Newton's rings and deduce an expression for the diameter of bright and dark rings.	1	3	10
		OR			
3.		Explain the Fraunhofer diffraction due to single slit and obtain the conditions for principal maximum, minima, and secondary maxima.	1	3	10
		UNIT-2			
4.	**a)**	Describe the construction and working of a He-Ne laser.	2	3	5
	b)	Distinguish spontaneous and stimulated emissions.	2	3	5
		OR			
5.	**a)**	Classify the types of optical fibers based on their refractive index profile.	2	4	5
	b)	Derive the expression for acceptance angle? Discuss its relation with Numerical aperture.	2	3	5
		UNIT-3			
6.	**a)**	Derive the expression for electrical conductivity based on quantum free electron theory.	3	3	5
	b)	Distinguish between classical free electron theory and quantum free electron theory of metals.	3	3	5
		OR			

7.	**a)**	Explain the phenomenon of Hall effect? Derive the expression for Hall coefficient.	3	3	5
	b)	The Hall coefficient of a specimen is $3.66 \times 10^{-4}\ m^3C^{-1}$. Its resistivity is $8.93\times 10^{-3}\Omega m$. Find carrier density.	3	3	5
		UNIT-4			
8.		Apply Schrodinger's wave equation to the case of a particle in a one dimensional potential well and show that the energies of the particle are quantized.	4	4	10
		OR			
9.	**a)**	Distinguish between classical bits and qubits	4	3	5
	b)	Explain the basic principles of quantum computation.	4	3	5
		UNIT-5			
10.	**a).**	Apply Gauss divergence theorem to derive the differential form of Gauss law for the electric field.	5	3	5
	b).	State Stokes theorem and explain its physical significance.	5	3	5
		OR			
11.		Derive Maxwell's equations in electro statics and magneto statics in integral form and mention their significance.	5	3	10

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610005
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C			
		(Common to AIML, CIC, CSD, CSE, CSIT, ECE, IT)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Explain the rules for naming identifiers in C with suitable examples.	**1**	**2**	**2**
	b).	Differentiate between break and continue statements.	**2**	**2**	**2**
	c).	Explain the concept of a union in C with a suitable example.	**3**	**2**	**2**
	d).	Explain the concept of a pointer in C with a suitable example.	**4**	**2**	**2**
	e).	Explain how the #define directive is used to create a macro with an example.	**5**	**2**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Explain the concept of Computational Thinking and its main components with a suitable example.	**1**	**2**	**5**
	b).	Explain the different data types in C with suitable examples.	**1**	**2**	**5**
		OR			
3.	**a).**	Explain the methods used to convert a decimal number into binary, octal, and hexadecimal number systems with suitable examples.	**1**	**2**	**5**
	b).	Explain the different types of operators in C with suitable examples.	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Implement a C program using the switch statement to create a menu-driven calculator that performs addition, subtraction, multiplication, and division.	**2**	**3**	**5**
	b).	Explain the different looping statements in C with suitable examples.	**2**	**2**	**5**
		OR			
5.	**a).**	Explain the different types of if statements in C with suitable examples.	**2**	**2**	**5**
	b).	Implement a C program to sort a given array of elements in ascending order using the Bubble Sort technique.	**2**	**3**	**5**
		UNIT-3			

6.	**a).**	Explain the commonly used string handling functions in C with their syntax and suitable examples.	**3**	**2**	**5**
	b).	Implement a C program using a structure named Employee to store employee ID, name, and salary, and display the employee details.	**3**	**3**	**5**
		OR			
7.	**a).**	Implement a C program to read a string and count the number of vowels, consonants, digits, and special characters in the given string.	**3**	**3**	**5**
	b).	Explain typedef and enumerated data types in C with their syntax and suitable examples.	**3**	**2**	**5**
		UNIT-4			
8.	**a).**	Explain pointers in C, including pointer declaration, initialization, and accessing addresses and values, with suitable examples.	**4**	**2**	**5**
	b).	Illustrate the parameter passing techniques in C with suitable examples.	**4**	**3**	**5**
		OR			
9.	**a).**	Explain dynamic memory allocation in C using malloc() and calloc() with suitable examples.	**4**	**2**	**5**
	b).	Implement a C program to calculate the factorial of a given number n using recursion.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain preprocessor directives and macros in C with suitable examples.	**5**	**2**	**5**
	b).	Implement a C program using file handling functions to create, store, and display student data in a file.	**5**	**3**	**5**
		OR			
11.	**a).**	Explain the commonly used file handling functions in C with their syntax and suitable examples.	**5**	**2**	**5**
	b).	Implement a C program to copy the contents of a source file into a destination file using file handling functions.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610402
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		ELEMENTS OF ELECTRICAL ENGINEERING			
		(For ECE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Define RMS value and frequency of a sinusoidal voltage.	**1**	**2**	**2**
	b).	List any four major equipment used in a substation.	**2**	**1**	**2**
	c).	State any two applications of a servo motor.	**3**	**1**	**2**
	d).	What is a rectifier?	**4**	**2**	**2**
	e).	What is electric shock?	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Illustrate the principle of operation of a 3-phase alternator with a neat sketch.	**1**	**3**	**5**
	b).	Compare conventional and non-conventional sources of electrical energy.	**1**	**3**	**5**
		OR			
3.	**a).**	Explain the principle of solar power generation and state its advantages.	**1**	**3**	**5**
	b).	Illustrate the working of a hydroelectric power plant with a neat schematic diagram.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Illustrate the construction and working principle of a transformer with a neat sketch.	**2**	**3**	**5**
	b).	Explain the radial distribution system with a neat schematic diagram.	**2**	**3**	**5**
		OR			
5.	**a).**	Explain the layout of a substation and describe its major equipment.	**2**	**3**	**5**
	b).	Explain the ring-main distribution system with a neat schematic diagram.	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	Derive and explain the torque equation of a DC motor.	**3**	**3**	**5**
	b).	Explain the construction and principle of operation of a BLDC motor.	**3**	**3**	**5**
		OR			

7.	**a).**	Apply the concept of slip to determine its value in a three-phase induction motor and state its significance.	**3**	**3**	**5**
	b).	Explain the construction and principle of operation of a stepper motor.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Explain the working of a single-phase full-wave diode bridge rectifier with a neat circuit diagram.	**4**	**3**	**5**
	b).	Select the different types of suitable batteries and their applications.	**4**	**3**	**5**
		OR			
9.	**a).**	Illustrate the construction and working of a lead-acid battery.	**4**	**3**	**5**
	b).	Apply the concepts of SOC and DOD to determine the operating condition of a battery and illustrate the basic arrangement of a battery pack and UPS.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain the different types of hazards in electrical systems.	**5**	**3**	**5**
	b).	Explain the purpose and importance of earthing in electrical installations.	**5**	**3**	**5**
		OR			
11.	**a).**	Apply electrical safety principles to identify the causes and effects of electric shock and recommend suitable preventive measures.	**5**	**3**	**5**
	b).	Explain the construction, working and applications of an MCB.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2620001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Find the integrating factor of $\frac{dy}{dx} - \frac{3y}{x} = x$	**1**	**3**	**2**
	b).	Solve$(D^2 - 4D + 4)y = 0$	**2**	**3**	**2**
	c).	Obtain a P.D.E from $z = ax + by + c$	**3**	**2**	**2**
	d).	Determine $\nabla\emptyset$, where $\emptyset = \sqrt{x^2 + y^2 + z^2}$	**4**	**3**	**2**
	e).	Find div $\bar{F}$, where $\bar{F} = x\bar{\imath} + y\bar{\jmath} + z\bar{k}$	**5**	**3**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Solve $y \log y \, dx + (x - \log y)dy = 0$	**1**	**3**	**5**
	b).	Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$	**1**	**3**	**5**
		OR			
3.	**a).**	If the air is maintained at 30ºC and the temperature of the body cools from 100ºC to 70º C in 15 minutes, determine the time at which the temperature of the body will be 40º C.	**1**	**3**	**5**
	b).	Find the orthogonal trajectories of the family of curves $r^2 = a^2 Sin2\theta$	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Solve $(D-2)^2 y = 8(e^{2x} + x^2)$	**2**	**3**	**5**
	b).	Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + y = \tan x$	**2**	**3**	**5**
		OR			
5.	**a).**	Solve simultaneous equations $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that x = 2 and y=0 when t =0	**2**	**3**	**5**
	b).	The damped LCR circuit is governed by the differential equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$ where L, C, R are positive constants. Find the critical resistance.	**2**	**3**	**5**

		UNIT-3			
6.	**a).**	Form the partial differential equation by eliminating arbitrary function from $f(x^2+y^2+z^2, z^2-2xy)=0$.	**3**	**3**	**5**
	b).	Solve $x^2(y-z)p+y^2(z-x)q=z^2(x-y)$	**3**	**3**	**5**
		OR			
7.		Solve $\left(D^2-3DD'+2D'^2\right)z=e^{x+2y}+Sin(x+2y)$	**3**	**3**	**10**
		UNIT-4			
8.	**a).**	Find the directional derivative of $f = xy+yz+zx$ in the direction of vector $\bar{i}+2\bar{j}+2\bar{k}$ at the point (1,2,0).	**4**	**3**	**5**
	b).	If $\overrightarrow{F} = 3x^2\overrightarrow{\imath}+5xy^2\overrightarrow{j}+5xyz^3\vec{k}$,then find $\nabla(\nabla.\vec{F})$	**4**	**3**	**5**
		OR			
9.	**a).**	Prove that $(y^2-z^2+3yz-2x)\vec{\imath}+(3xz+2xy)\vec{j}+(3xy-2xz+2z)\vec{k}$ is solenoidal.	**4**	**3**	**5**
	b).	Find the angle between the surfaces $xy^2z=3x+z^2$ and $3x^2-y^2+2z=1$ at the point (1,–2,1)	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Find the work done by the force field $\vec{F}=(2x+y^2)\vec{\imath}+xz\vec{j}+xyz\vec{k}$ in moving a particle along a straight line from (0,0,0) to (2,1,3).	**5**	**3**	**5**
	b).	Evaluate $\oint_c\{(x^2y\,dx+x^2dy\}$ where C is a circle $x^2+y^2=a^2$ by using Green's theorem.	**5**	**3**	**5**
		OR			
11.		Verify Gauss – divergence theorem for $\vec{F}=2xz\vec{\imath}+yz\vec{j}+z^2\vec{k}$ where S is an upper half surface of the sphere $x^2+y^2+z^2=a^2$	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

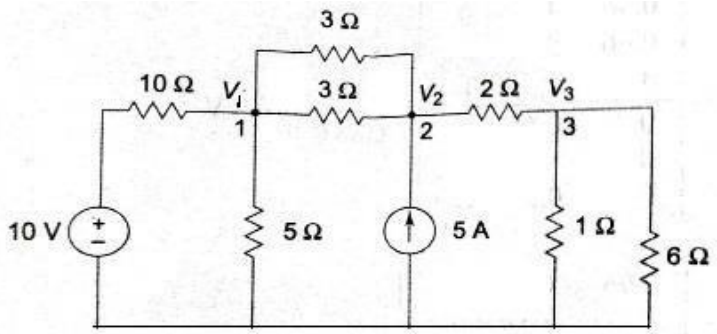
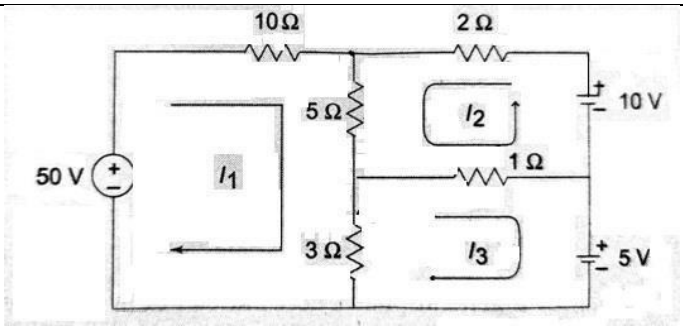
NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

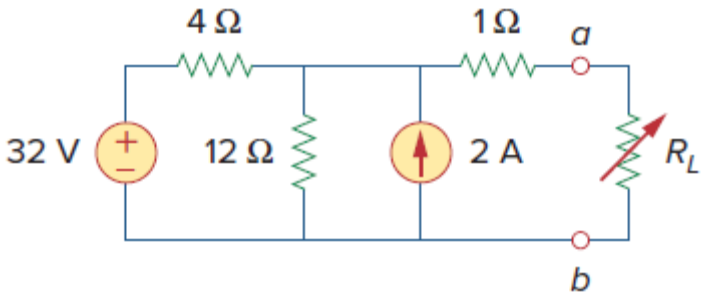
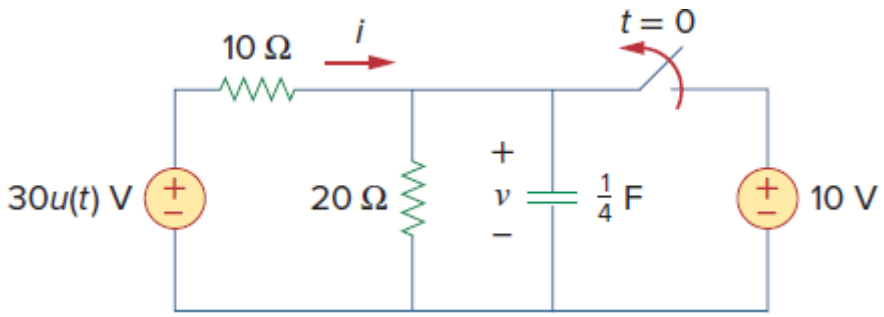
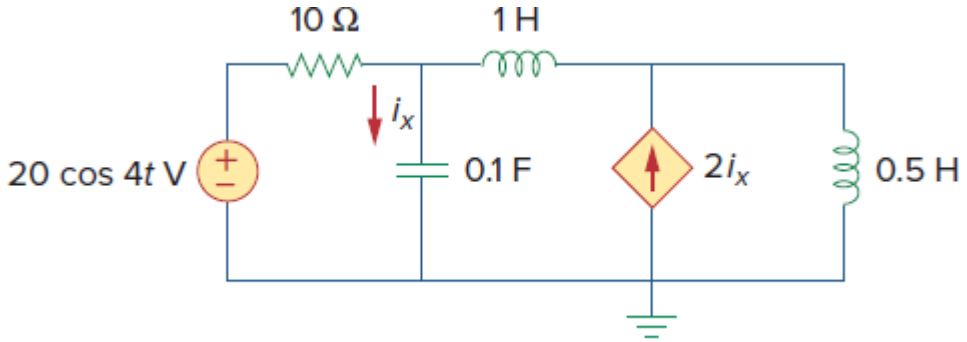
					Course Code:A2620401
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		APPLIED CHEMISRTY			
		(For ECE)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	How does an increase in the frequency of incident radiation affect the kinetic energy of photoelectrons?	**1**	**3**	**2**
	b).	How can the electrochemical series be used to predict the feasibility of a redox reaction?	**2**	**3**	**2**
	c).	Why is SiC preferred for high-temperature and high-power electronic applications?	**3**	**3**	**2**
	d).	Distinguish between stretching and bending vibrations.	**4**	**3**	**2**
	e).	Mention two applications of graphene.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Analyze the wave-particle duality of matter with reference to de Broglie's hypothesis.	**1**	**4**	**5**
	b).	Using the Schrödinger wave equation, obtain the normalized wave function for a particle in a one-dimensional box of length *L*.	**1**	**3**	**5**
		OR			
3.	**a).**	Analyze the molecular orbital energy-level diagrams of O_2 and N_2 and explain the difference in their magnetic properties.	**1**	**4**	**5**
	b).	Using π-MO theory, explain the formation of π-molecular orbitals in benzene and identify the bonding and antibonding molecular orbitals.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Describe how an ion-selective electrode can be used for the determination of a specific ion in solution.	**2**	**3**	**5**
	b).	Interpret the conductance variation during the strong acid–strong base conductometric titration and explain the shape of the titration curve.	**2**	**4**	**5**
		OR			

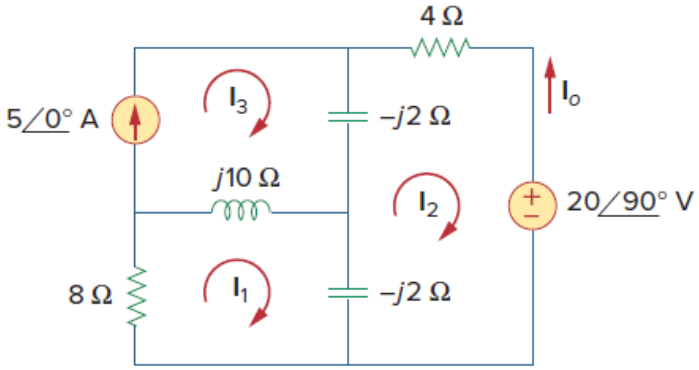
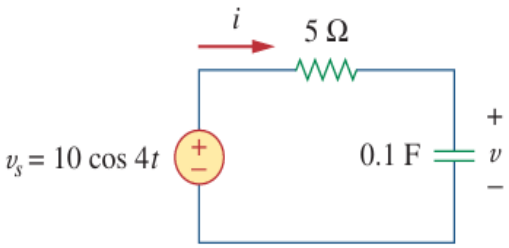
5.	**a).**	Explain how capacity, power density and cycle life influence the selection of a battery for a particular application.	**2**	**3**	**5**
	b).	Compare lead-acid and lithium-ion batteries with respect to construction, working and applications.	**2**	**4**	**5**
		UNIT-3			
6.	**a).**	Discuss the conduction mechanism of polyacetylene and explain the role of conjugation in enhancing its conductivity.	**3**	**4**	**5**
	b).	Describe the role of perovskite semiconductor materials in modern photovoltaic devices.	**3**	**3**	**5**
		OR			
7.	**a).**	Relate the thermosetting nature and properties of Bakelite to its applications in electrical and engineering components.	**3**	**3**	**5**
	b).	Compare polyimide, BCB and PBO based on their applications in microelectronics.	**3**	**4**	**5**
		UNIT-4			
8.	**a).**	Compare amperometric and potentiometric sensors based on their working principle and applications.	4	4	**5**
	b).	Describe the principle and instrumentation of a colorimeter and explain its application in the estimation of iron	**4**	**3**	**5**
		OR			
9.	**a).**	Differentiate between the electronic transitions involved in UV–Visible spectroscopy based on their energy requirements.	**4**	**4**	**5**
	b).	Compare the applications of colorimetry, UV–Visible spectroscopy, and IR spectroscopy in chemical analysis.	**4**	**4**	**5**
		UNIT-5			
10.	**a).**	Compare the CVD and arc-discharge methods for the synthesis of carbon nanotubes.	**5**	**4**	**5**
	b).	Explain the biological method for the preparation of metallic nanoparticles and mention its advantages.	**5**	**3**	**5**
		OR			
11.	**a).**	Describe how nanomaterials contribute to the development of smaller, faster, and efficient electronic and memory devices.	**5**	**3**	**5**
	b).	Describe the basic concept of topological insulators and mention their applications in spintronics.	**5**	**3**	**5**

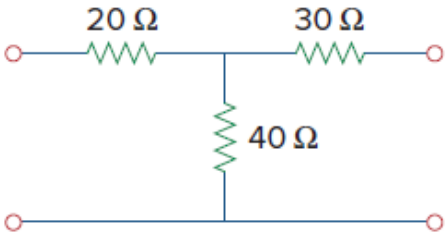
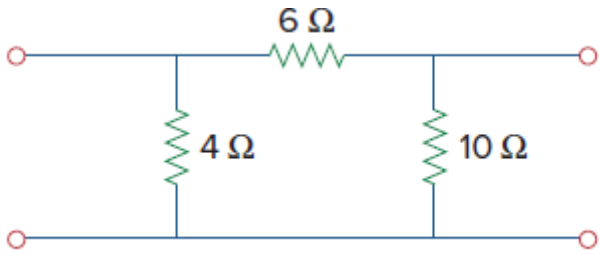
CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620403
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		NETWORK ANALYSIS			
		For ECE			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Explain KVL and KCL.	**1**	**1**	**2**
	b).	Explain Time Constant of a series RL circuit.	**2**	**2**	**2**
	c).	Define Impedance and Phase angle.	**3**	**1**	**2**
	d).	What is Coefficient of Coupling.	**4**	**2**	**2**
	e).	Define Image impedance and Iterative impedance.	**5**	**1**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Determine the voltages at each node for the circuit shown below using nodal analysis. 	**1**	**4**	**5**
	b).	Determine the mesh current i1, i2, i3 in the circuit using mesh analysis	**1**	**4**	**5**
					
		OR			

3.	**a).**	State Thevenin's theorem and find the Thevenin's equivalent V_{th} and R_{th} for the following circuit: 	**1**	**3**	**5**
	b).	State and prove Maximum power transfer theorem.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Find the step response of RC circuit.	**2**	**3**	**5**
	b).	In the circuit shown below, the switch has been closed for a long time and is opened at t=0. find i and v values for all time. 	**2**	**3**	**5**
		OR			
5.	**a).**	Obtain an expression for step response of series RL circuit.	**2**	**3**	**5**
	b).	State the properties of Laplace Transform.	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	Find the currents i_X in the circuit using nodal analysis 	**3**	**4**	**5**

	b).	Determine the current I_o in the circuit using mesh analysis. 	**3**	**4**	**5**
		OR			
7.	**a).**	Explain Series RL, RC, RLC Circuits.	**3**	**2**	**5**
	b).	Find v(t) and i(t) in the given circuit. 	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Explain Series Resonance.	**4**	**2**	**5**
	b).	Explain Parallel Resonance.	**4**	**2**	**5**
		OR			
9.	**a).**	Explain self-inductance and mutual inductance.	**4**	**2**	**5**
	b).	Derive an expression for Coefficient of coupling.	**4**	**2**	**5**
		UNIT-5			
10.	**a).**	Relate Y parameters and Z parameters.	**5**	**2**	**5**
	b).	Explain Cascade connection of Two Port networks.	**5**	**2**	**5**
		OR			
11.	**a).**	Find Z parameters of the network shown below	**5**	**3**	**5**

					
	b).	Determine the hybrid parameters of the network shown below. 	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

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